

Dugong Watch

Seagrass Habitat Risk Mapping for Dugong Conservation

Marawah / Bu Tinah Marine Biosphere Reserve — Abu Dhabi, UAE

Prototypes for Humanity 2026 — Submission

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Executive Summary

A dugong can live for seventy years and spend almost all of them grazing seagrass in shallow water. The Arabian Gulf holds the world's second-largest dugong population (~3,000–3,500), concentrated in Abu Dhabi's UNESCO-designated Marawah / Bu Tinah reserve — yet the seagrass they depend on is under pressure from coastal development, dredging, desalination discharge, marine heat, and vessel traffic.

Dugong Watch answers a question conservation planners currently cannot: not “where is there seagrass,” but *where is seagrass both present and under enough pressure that its loss would matter for dugongs* — so limited protection resources can be aimed first. It is an end-to-end, open pipeline built on free satellite data: a Random Forest classifier maps seagrass from Sentinel-2 at **85.3% accuracy**, change is detected between 2019 and 2024, and habitat value is combined with five weighted threat factors into one risk index ($\text{Risk} = \text{Value} \times \text{Threat}$), yielding **105 ranked hotspots** delivered through an interactive dashboard.

The output is validated three ways: risk falls with distance from real pressure, is **3× higher inside the documented dugong core** than reserve-wide, and shifts by $\leq 3\%$ when any weight is perturbed $\pm 10\%$. Every input is free and public and every step is code, so it re-runs at near-zero cost as new imagery arrives — and the same structure transfers to other satellite-visible, single-species habitats worldwide.

1. Introduction & Problem Statement

1.1 Background

The dugong (*Dugong dugon*) feeds almost exclusively on seagrass: protect the meadows and you protect the animal; lose them and it has nowhere left to feed. The Arabian Gulf population, centred on Abu Dhabi, is the world's second-largest after Australia's (~3,000–3,500 individuals), and the Marawah Marine Biosphere Reserve holds its highest concentration.

That seagrass is under measurable pressure: coastal development and dredging, desalination and industrial discharge, marine heatwaves in one of the hottest seas on Earth, and vessel traffic in the shallow (<5 m) waters where dugongs feed. Planning in the region therefore needs a specific answer — not “where is there seagrass,” but **where is seagrass both present and under enough pressure that its loss would matter for dugong survival.**

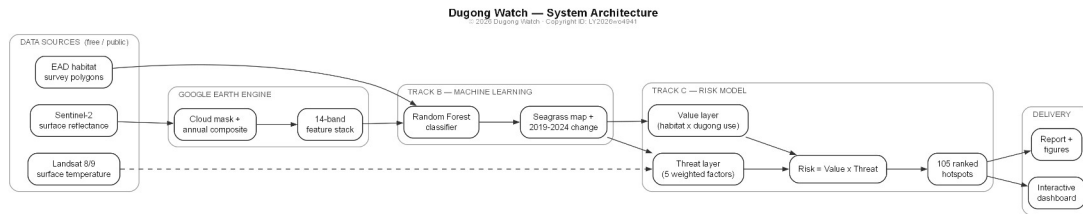
1.2 Problem statement

Answering that means combining three things rarely brought together: habitat maps (where seagrass is), habitat change (whether it is being lost), and threat data (what pressures act where). Habitat surveys are episodic and expensive; threat data is scattered and non-standardised; and there is no publicly available, reproducible model that fuses these into a ranked map of *risk to dugong habitat specifically* — as opposed to generic seagrass extent maps or generic risk indices not conditioned on dugong presence. What planners actually need is a transparent, updatable answer to “if we can only act in a few places, where first?”

1.3 Project objective

Dugong Watch is an end-to-end, reproducible pipeline that (1) classifies seagrass from free Sentinel-2 imagery with a model trained and validated on official EAD habitat surveys; (2) detects seagrass change between two periods; (3) combines habitat with five geographically specific threat factors — observed loss, coastal development, thermal stress, desalination discharge, and vessel pressure — into one weighted, explicitly justified risk index; and (4) surfaces the result as a ranked, interactive map, alongside the code and data so every step can be audited or reproduced.

The guiding principle is **transparency over sophistication**: every weight is a documented parameter, every threat layer's provenance is labelled, and the model's sensitivity to its assumptions is tested, not assumed.



System architecture: free satellite imagery and official survey data flow through Google Earth Engine into a machine-learning classifier and a weighted risk model, producing a ranked, auditable output delivered as an interactive dashboard and report.

1.4 Scope

The project covers the Marawah / Bu Tinah region of the Arabian Gulf, comparing 2019 and 2024 (the best cloud-free Sentinel-2 coverage). Classification is binary (1 = seagrass, 0 = non-seagrass), matching the EAD training data. The risk index is explicitly **relative** — it ranks locations within the study area against each other, not against an absolute probability of habitat loss.

2. Literature Review

2.1 Dugong ecology and regional context

Marsh et al. (2002, 2011) established the baseline understanding of dugong ecology and the threats driving decline — habitat loss, bycatch, and vessel strike foremost among them. This project weights observed habitat loss highest precisely because the literature treats it as the most consequential and most directly observable driver. The Marine Mammal Protected Areas Task Force’s *Southern Gulf IMMA* factsheet documents Abu Dhabi’s population as the second-largest globally, with the Marawah / Bu Tinah area at highest regional density — the basis for this project’s “dugong use” surface (Section 3.3). The EAD’s 2015 aerial dugong survey corroborates that pattern, and the Dugong & Seagrass Conservation Project (UAE) documents land reclamation and dredging as the leading local drivers of seagrass loss, and that Gulf meadows support only three heat-tolerant seagrass species under summer sea-surface temperatures regularly exceeding 36 °C — both incorporated directly into the threat weighting.

2.2 Habitat risk assessment methodology

The core framework — risk as **exposure × consequence**, high only where valuable habitat *and* real pressure coincide — follows two established sources. Halpern et al. (2008) established the cumulative human-impact mapping approach behind this project’s weighted threat overlay. Arkema et al. (2013) and the InVEST Habitat Risk Assessment (Natural Capital Project) formalise the exposure-times-consequence logic that $\text{Risk} = \text{Value} \times \text{Threat}$ implements, including normalising each factor to a common 0–1 scale before combination. This is why the index multiplies value by threat rather than summing threats alone — which would flag high-pressure but seagrass-free water as “risky,” a meaningless output for conservation planning.

2.3 Regional and institutional context

The UNESCO Man and the Biosphere designation of the Marawah Reserve (2007), and its Marine Protected Area status since 2001, make it the largest MPA in the Arabian Gulf (4,255 km²) and the institutional context for the study area — and the reason habitat inside the reserve, and especially the Bu Tinah no-take core, carries elevated weight in the model.

2.4 Gap this project addresses

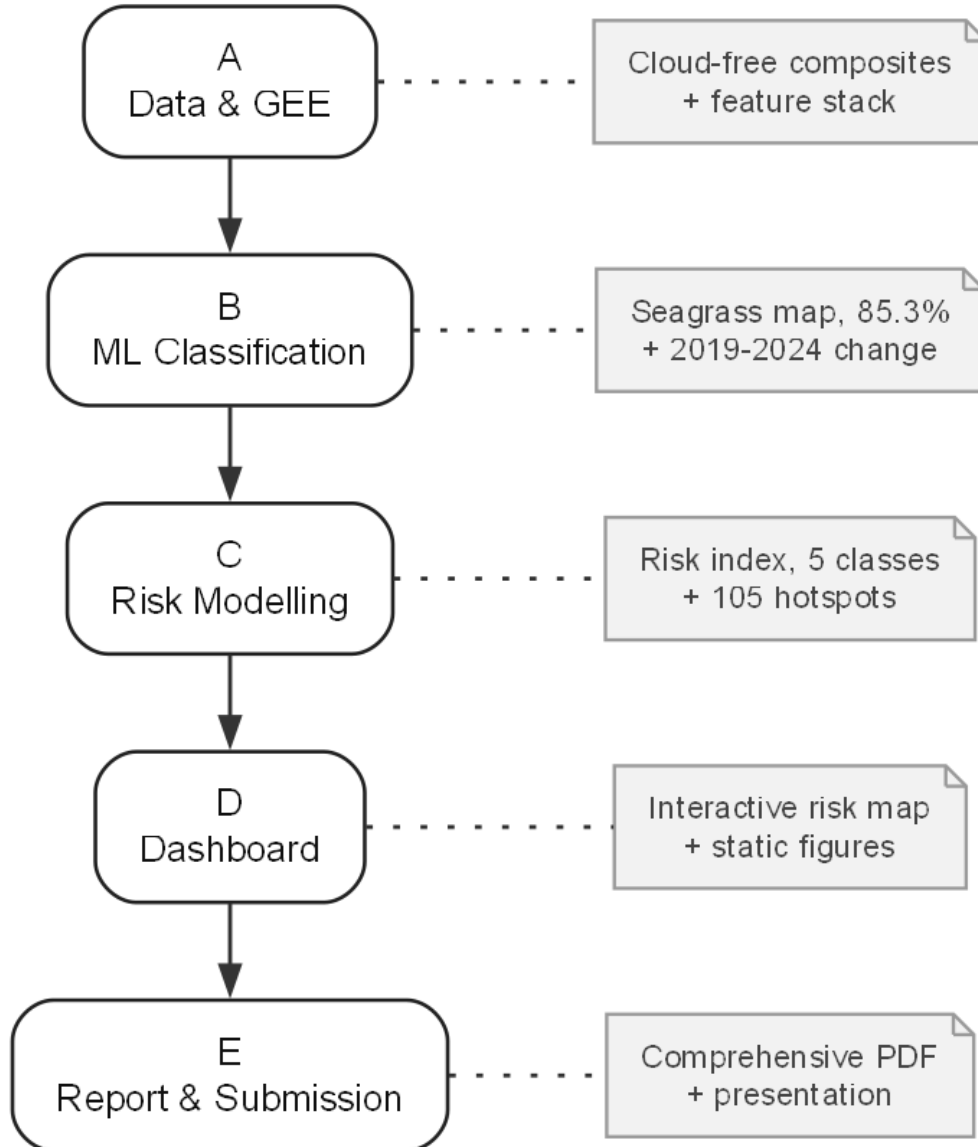
None of these sources combine seagrass classification, observed change, and multi-factor threat data into a single, ranked, reproducible risk output for dugong habitat in this region. They describe *where dugongs are*, *what threatens them*, and *how to combine exposure and threat in general*; this project applies that general framework with region-specific data and justified weights to produce a specific, auditable output for Marawah / Bu Tinah.

3. Methodology

Every number below is drawn directly from the project's two notebooks (M2_classification/Seagrass_Dugong_Mapping.ipynb for classification and C_risk_gis/Dugong_Habitat_Risk.ipynb for the risk model), both run against live data and verified.

Dugong Watch — Five-Track Pipeline

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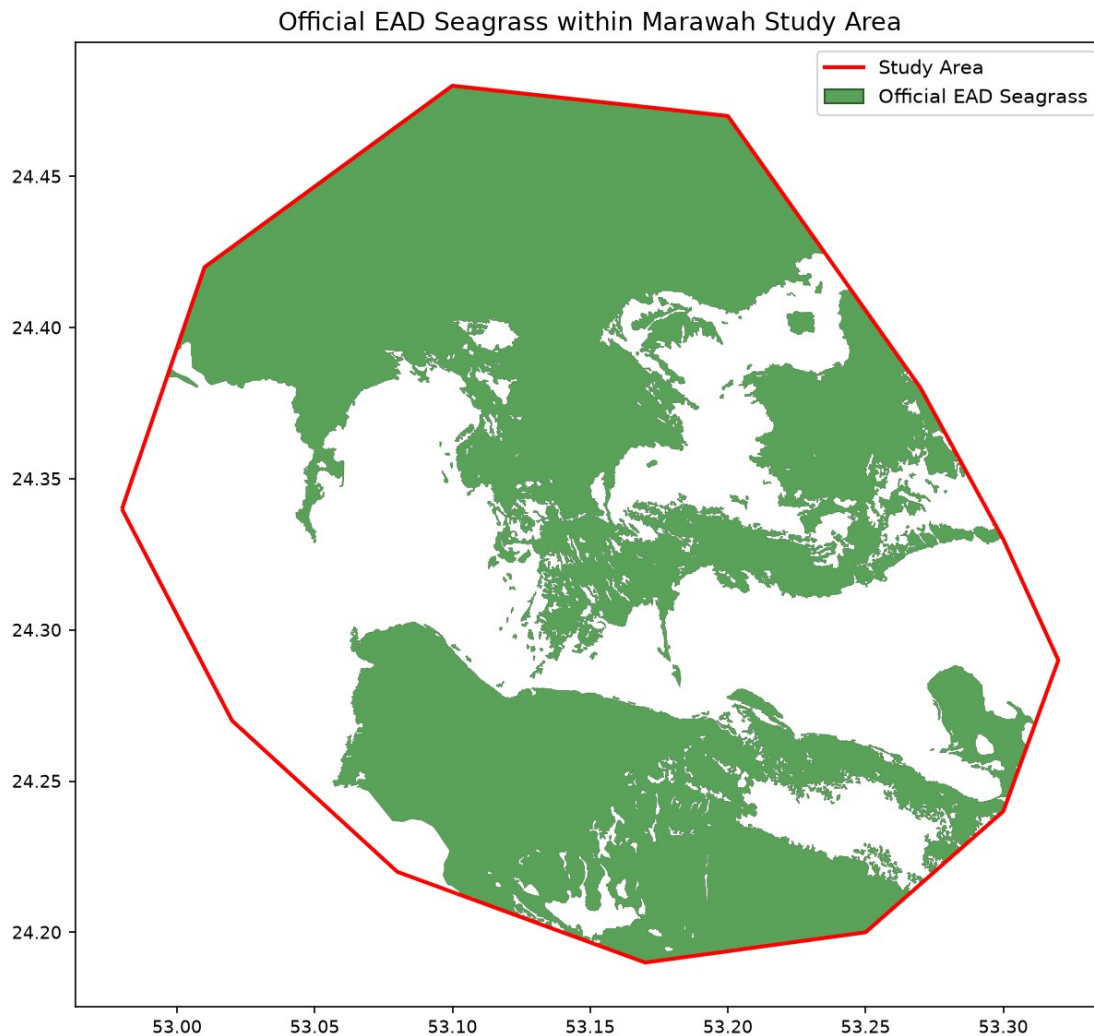
The five-track pipeline: each track (A–E) hands its output to the next, from data acquisition through machine learning, risk modelling, the dashboard, and this report.

3.1 Study area and data

The study area is a 774.6 km² polygon over the Marawah / Bu Tinah region of the Arabian Gulf (lon 52.98–53.32, lat 24.19–24.48), entirely inside the Marawah Marine Biosphere Reserve.

- **Satellite imagery:** Sentinel-2 Surface Reflectance (COPERNICUS/S2_SR_HARMONIZED) via Google Earth Engine, cloud-masked (QA60) and median-composited per year; 222 scenes built the 2024 composite. Composites for 2019 and 2024 are compared.
- **Thermal data:** Landsat 8/9 summer surface temperature (ST_B10, June–September), converted to Celsius and water-masked so hot dry land cannot skew the signal.
- **Training and reference data (EAD):** 118 seagrass and 171 non-seagrass polygons for training; a separate 118-polygon seagrass layer for independent validation; and a 1,954-polygon regional layer for context only.

The binary scheme (1 = seagrass, 0 = non-seagrass) matches the EAD training data.



Official EAD seagrass reference beds (118 polygons) within the 774.6 km² study area — the ground truth the classifier is trained and validated against.

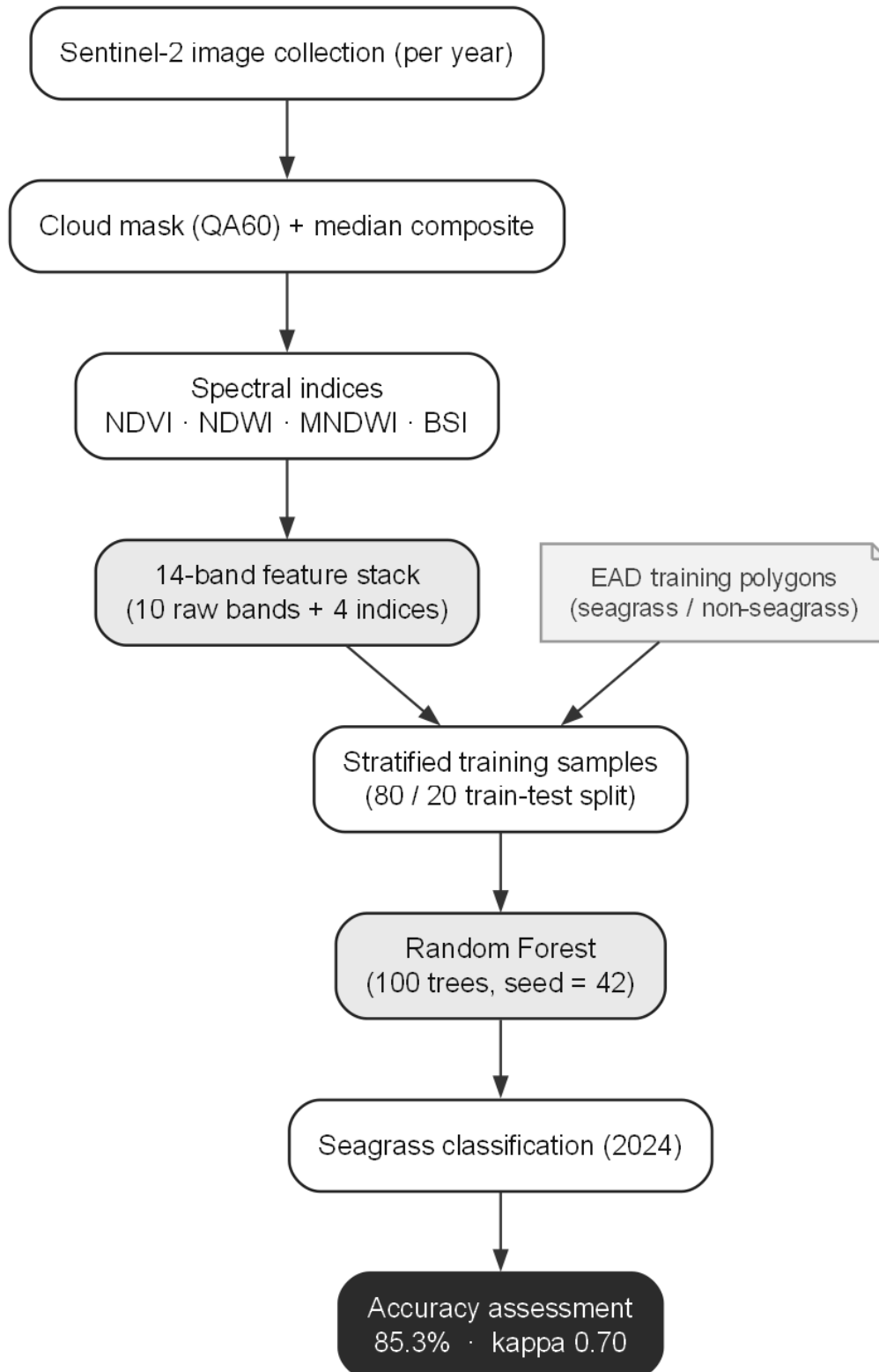
3.2 Seagrass classification (Track B)

Each composite is reduced to **14 predictor bands** — ten raw Sentinel-2 bands (B2–B12) plus four spectral indices (NDVI, NDWI, MNDWI, BSI). Stratified random points inside the

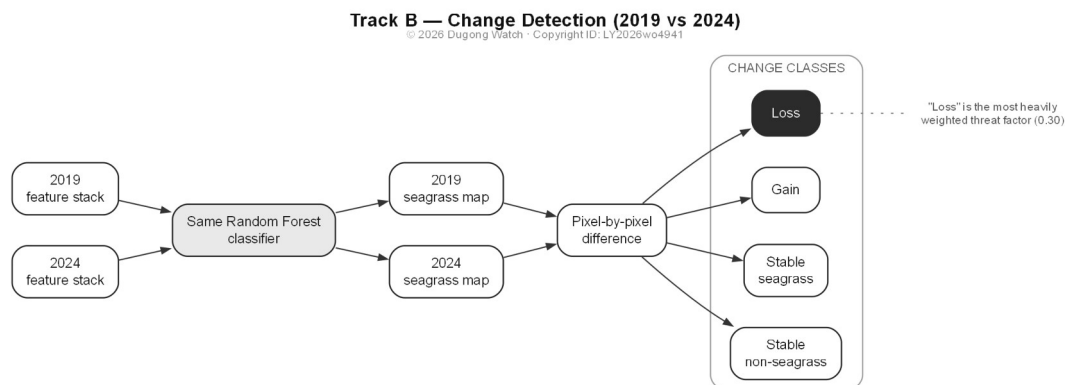
training polygons (not one per polygon, since sizes vary up to ~61 km²) supply each point's 14 band values at 10 m. A **Random Forest** (ee.Classifier.smileRandomForest, 100 trees, fixed seed) is trained on an 80/20 split; a class-balanced version outperformed the imbalanced one on every metric (Section 4) and is used downstream. For **change detection**, the same trained classifier is applied to both 2019 and 2024 (not retrained per period, so differences reflect real change, not model drift), and the two maps are differenced into four classes: stable non-seagrass, stable seagrass, gain, and loss.

Track B — Seagrass Classification Workflow

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The Track B seagrass-classification workflow, from Sentinel-2 imagery through the 14-band feature stack and Random Forest classifier to a validated seagrass map.



Change detection: the same trained classifier is applied to the 2019 and 2024 composites and the two maps are differenced into four change classes — stable non-seagrass, stable seagrass, gain, and loss.

3.3 Habitat-risk model (Track C)

The model follows an **exposure × consequence** structure (Section 2.2): $\text{Risk} = \text{Value} \times \text{Threat}$, so a location scores high only when valuable habitat and real pressure coincide.

Value (what is at stake):

$$\text{Value} = \text{SeagrassPresence} \times (0.6 + 0.4 \times \text{DugongUse})$$

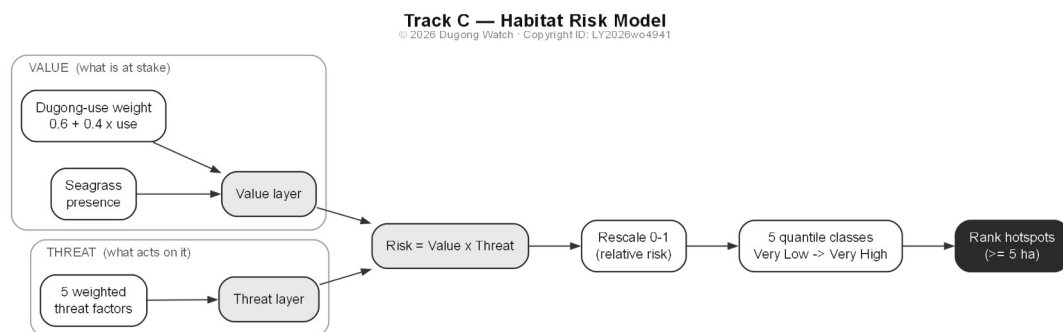
SeagrassPresence comes from the 2024 classification; DugongUse is a 0–1 distance-decay surface from documented high-density anchors (Marawah core, Bu Tinah, central foraging grounds), so core seagrass is weighted up to $\sim 1.7\times$ more than reserve-edge seagrass while every reserve pixel keeps some value.

Threat (what acts on it) — a weighted sum of five normalised (0–1) factors:

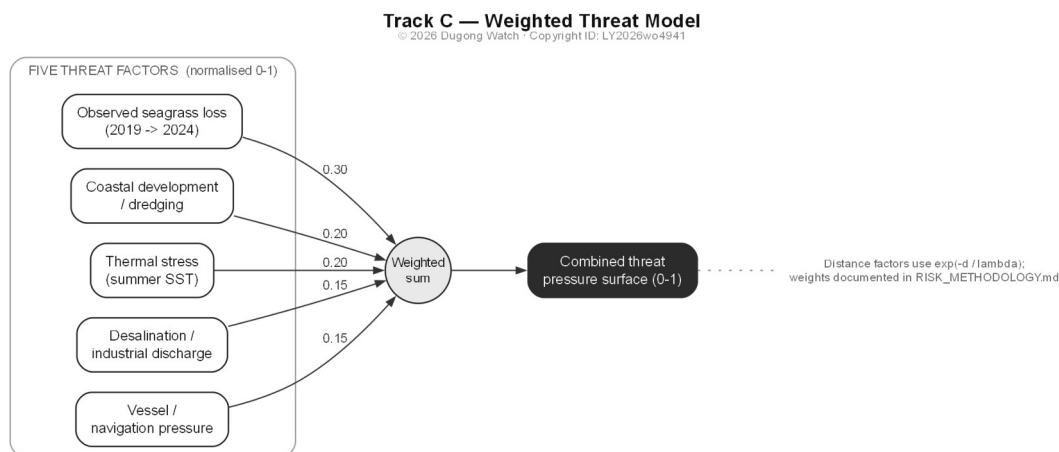
Factor	Measured as	Weight
Observed seagrass loss (2019 → 2024)	Smoothed density of “loss” change pixels	0.30
Coastal development / dredging	Distance decay from mapped development points	0.20
Thermal stress	Landsat summer surface temperature, water-masked	0.20
Desalination / industrial discharge	Distance decay from mapped discharge points	0.15

Factor	Measured as	Weight
Vessel / navigation pressure	Distance decay from ports, landings, routes	0.15

Distance factors use $\exp(-\text{distance}/\lambda)$ with factor-specific decay lengths ($\lambda \approx 3\text{--}5$ km), so distant regional plants contribute almost nothing while the nearer Mirfa plant registers on the reserve's south-east flank. Built- infrastructure coordinates are real; the reserve boundary, dugong-density surface, and navigation routes are documented approximations pending official EAD GIS layers. Risk = Value $\times$ Threat is rescaled 0–1 and binned into five classes (Very Low $\rightarrow$ Very High) using quantile breaks over seagrass pixels only.



The Track C habitat-risk model: a habitat-value layer multiplied by a weighted threat layer, rescaled, classified, and ranked into hotspots.



The weighted threat model: five normalised factors combined in documented proportions into a single threat-pressure surface.

3.4 Validation approach

Beyond standard accuracy metrics, three checks test whether the risk output is trustworthy rather than an artefact of its parameters (results in Section 4.3–4.4): (1) **face validity** — does risk fall

with distance from a real discharge point; (2) **zonal comparison** — is risk inside the dugong core higher than reserve-wide; and (3) **weight sensitivity** — do the hotspots survive perturbing each weight $\pm 10\%$.

3.5 Implementation and reproducibility

The whole pipeline runs in Google Earth Engine via Python (`earthengine-api`, `geemap`). Both notebooks re-run end-to-end using only the vector data committed to the repository plus Earth Engine’s public Sentinel-2 / Landsat archives — no proprietary or purchased data anywhere.

4. Results

All figures are generated directly from these results, not illustratively.

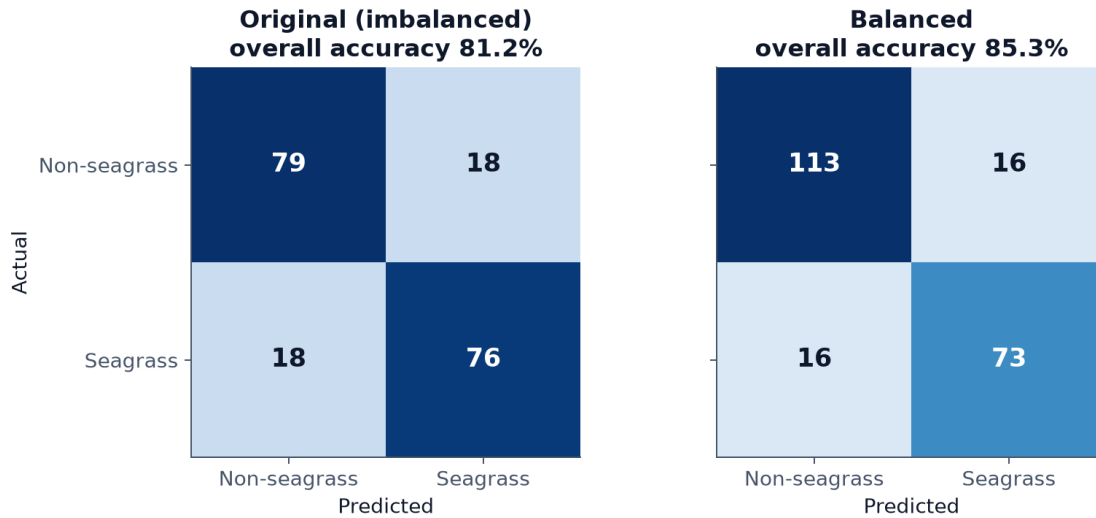
4.1 Seagrass classification accuracy

Two Random Forest models were compared (Section 3.2):

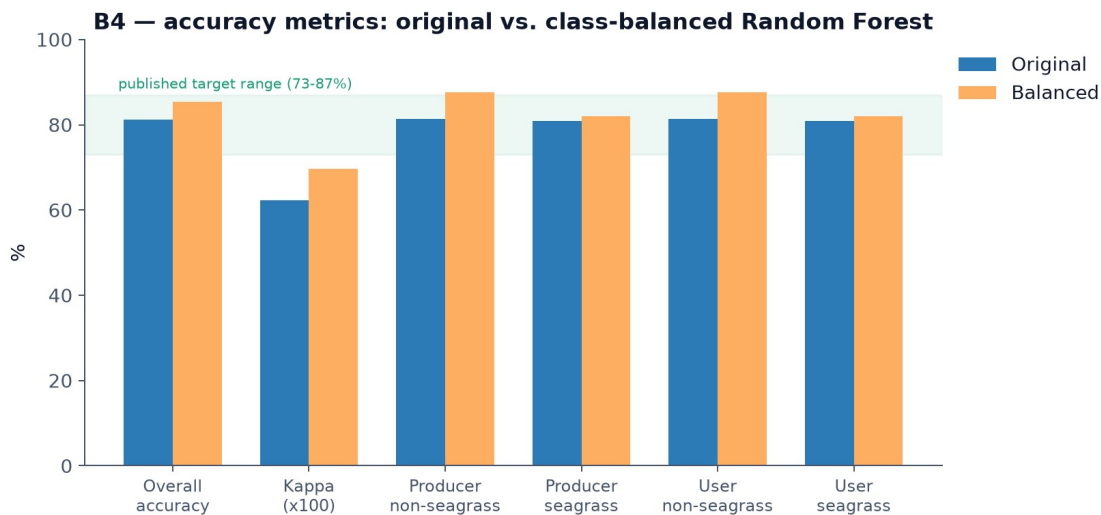
Metric	Original (imbalanced)	Class-balanced
Overall accuracy	81.15%	85.32%
Kappa	0.623	0.696
Producer’s accuracy (non-seagrass)	81.44%	87.60%
Producer’s accuracy (seagrass)	80.85%	82.02%
User’s accuracy (non-seagrass)	81.44%	87.60%
User’s accuracy (seagrass)	80.85%	82.02%

Both fall within the published 73–87% range for comparable satellite seagrass methods; the class-balanced model is used for all risk modelling.

B4 — Random Forest accuracy: seagrass classification (2024)

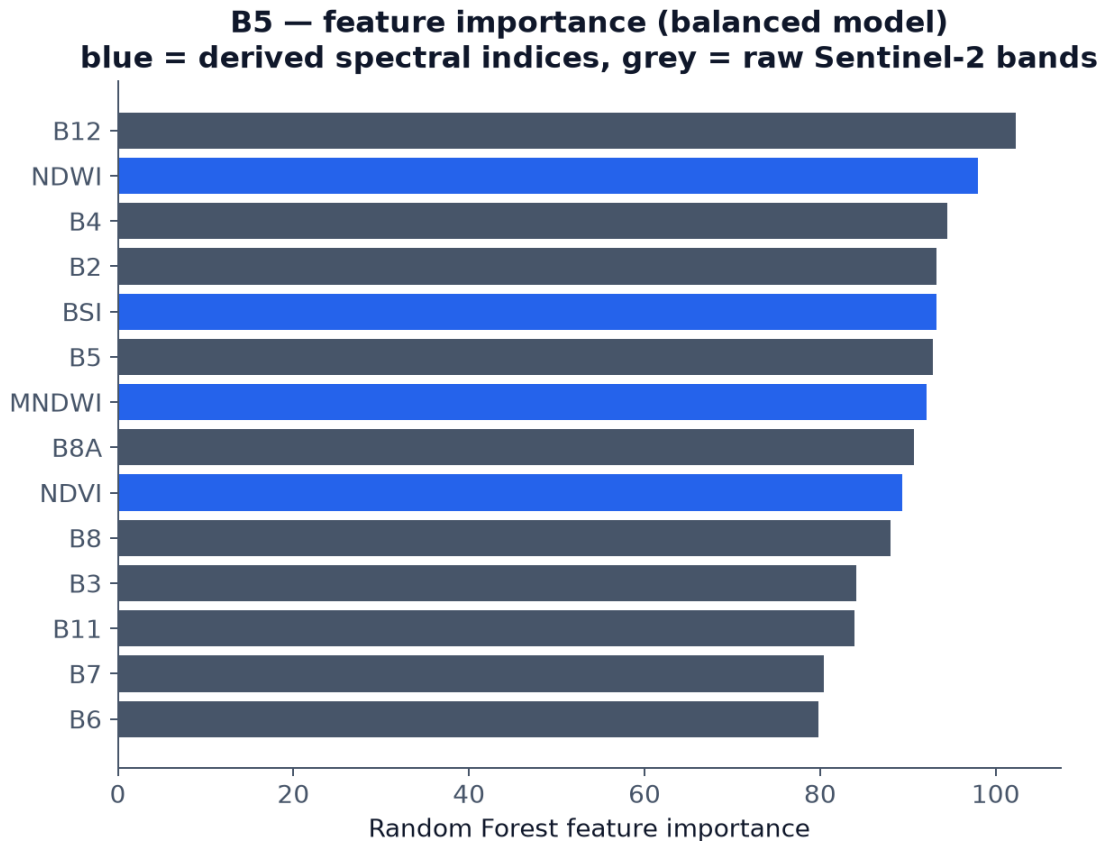


Confusion matrices: original vs. class-balanced Random Forest



Accuracy metrics compared against the published target range

Feature importance (balanced model, high to low): B12, NDWI, B4, B2, BSI, B5, MNDWI, B8A, NDVI, B8, B3, B11, B7, B6. Water- and vegetation-index bands (NDWI, BSI, MNDWI, NDVI) rank among the most important alongside the shortwave-infrared band B12 — consistent with seagrass discrimination depending on both spectral water characteristics and vegetation signal.

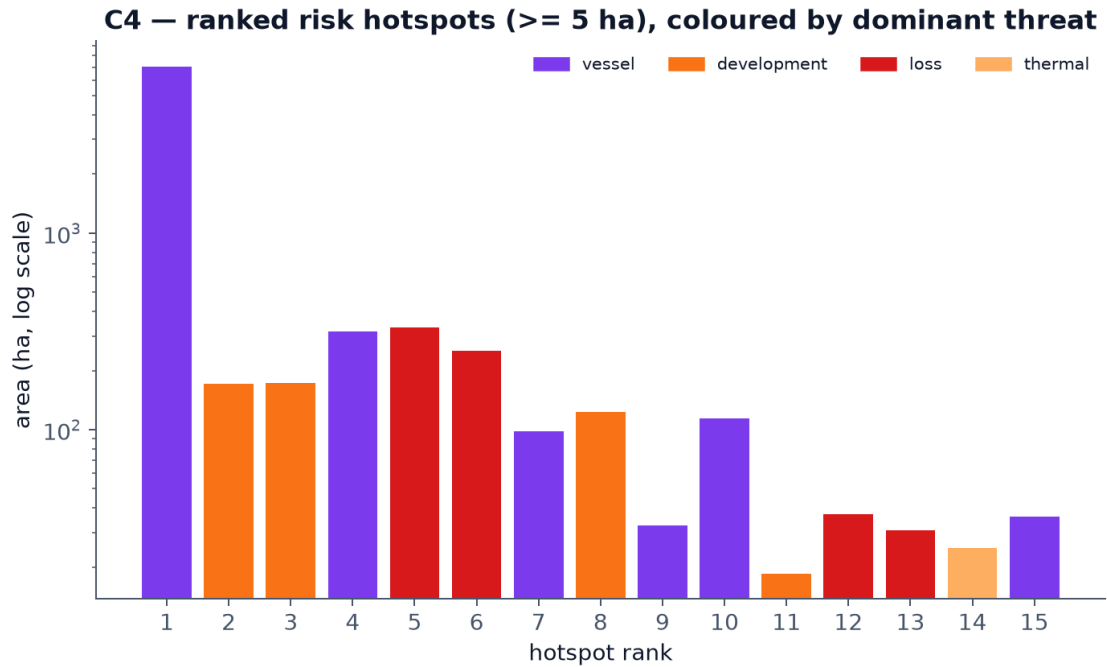


Random Forest feature importance

4.2 Habitat-risk index

Across the 774.6 km² study area the model produced:

- **Risk class breaks** (quantiles over seagrass pixels, 0–1 scale): 0.049 / 0.092 / 0.158 / 0.24, separating Very Low to Very High.
- **105 ranked hotspots** (≥ 5 ha, top class), from 18.5 to 7,048.9 ha, mean risk 0.293–0.659 across the top 15. The largest (7,048.9 ha, vessel-dominated) is one contiguous high-pressure zone rather than an artefact; smaller, higher-mean-risk hotspots are development- and loss-dominated.
- Dominant threat across the top 15: vessel pressure (6), coastal development (4), observed loss (4), thermal stress (1).



Ranked risk hotspots by area, coloured by dominant threat



The habitat-risk map across the study area (blue = Very Low → red = Very High), with ranked hotspot markers on the interactive dashboard. The red corridor through the centre is the vessel-dominated navigation channel.

4.3 Model validation

Face validity: mean risk fell from ~0.26 near the nearest discharge point (20 km band) to ~0.09–0.14 at 40–48 km — a real decline consistent with the model’s distance-decay construction.

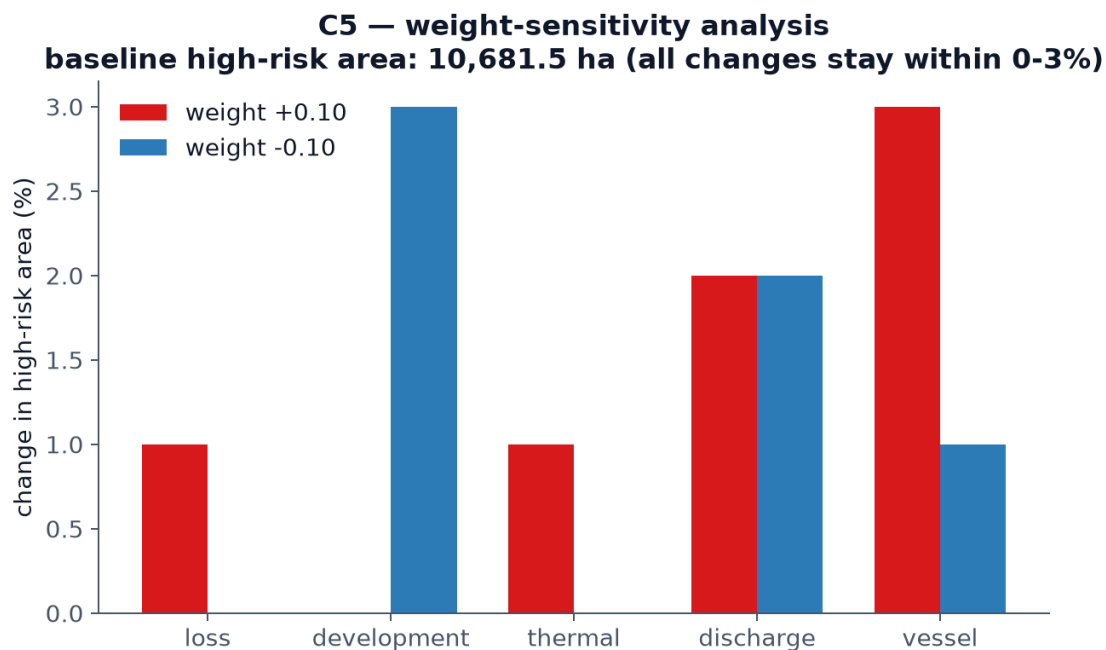
Zonal comparison: mean risk over seagrass inside the documented dugong core was **0.461** versus **0.153** reserve-wide — roughly **3× higher** in the core. This is the result a correct model should produce: risk concentrating where dugongs actually are, not uniformly or at random.

4.4 Weight-sensitivity analysis

Each weight was perturbed ± 0.10 (re-normalised to sum to 1.0) and the high-risk area recomputed:

Weight perturbed	+0.10	−0.10
Observed loss	+1%	+0%
Coastal development	−0%	+3%
Thermal stress	+1%	+0%
Discharge	+2%	+2%
Vessel pressure	+3%	+1%

Baseline high-risk area: 10,681.5 ha. No perturbation moved it by more than 3% — the ranked hotspots are robust to the exact weighting, not an artefact of one set of assumptions.



Weight-sensitivity analysis: change in high-risk area per ± 0.10 weight perturbation

4.5 Delivered outputs

Per run the pipeline exports a 5-class risk map, a continuous risk index, the underlying value and threat layers, a ranked hotspot vector layer, and the five threat context layers — all consumed directly by the interactive dashboard (Section 5).

5. Impact

5.1 Local — Abu Dhabi / UAE

Prioritised action. The 105 ranked hotspots answer a specific question for regulators: given limited enforcement, monitoring, or restoration budget, where first? Rather than a generic habitat map, they rank concrete locations by habitat value $\times$ real pressure — directly informing where a protected-area boundary review, a dredging-permit condition, or a monitoring buoy should go.

A monitoring baseline, not a snapshot. Built entirely on free, continuously updated Sentinel-2 / Landsat data, the pipeline re-runs at near-zero marginal cost; the 2019 $\rightarrow$ 2024 change layer already shows it can track whether hotspots stabilise, worsen, or respond to intervention.

Serving the reserve's mandate. The Marawah Reserve's UNESCO / MPA status commits it to evidence-based management; the 3 $\times$ zonal finding gives that mandate a concrete, quantified starting point rather than a general claim that “the core matters.”

5.2 Regional and global

The exposure $\times$ consequence structure, threat factors, and validation approach are not specific to seagrass or the Gulf: the same recipe — combine a habitat-value layer with a weighted, distance-decayed threat overlay, validate against ground truth, stress-test the weights — transfers to other single-species, satellite-visible marine habitats, from dugongs elsewhere in the Indo-Pacific to manatee seagrass in the Americas. Because it relies only on free imagery and open tooling, the barrier to adoption is know-how, not cost or licensing — relevant for lower-resource conservation programmes where commercial imagery or bespoke GIS consulting is not realistic.

5.3 Scope of these claims

Dugong Watch is a working, validated prototype. The impact above is what its outputs enable; realising it means putting the tool in front of the agencies who would use it and replacing schematic inputs with official GIS layers where available. Section 6 sets out these limitations in full — deliberately, because a tool that documents its own assumptions is more trustworthy to act on, not less.

6. Limitations

Stated directly, because transparency is part of the project's design (Section 1.3).

Data. Threat data is partially schematic: built-infrastructure coordinates are real, but the reserve boundary, dugong-density surface, and navigation routes are documented approximations

pending official EAD GIS layers — replacing them would sharpen results without changing the method. There is also a known training-data gap: a pagination bug truncated the non-seagrass training set to 171 of the true 319 polygons; the download is fixed, but the committed results were trained on the smaller set, so retraining on the full 319 is a specific, disclosed next step. Change detection depends on classifier accuracy at both time points, so loss-dominated hotspots should be read alongside the accuracy figures (Section 4.1). Thermal stress uses a summer 2020–2024 median — typical stress, not heatwave frequency.

Model. The index is **relative**, not a calibrated probability of loss. Weights are documented, literature-justified parameters shown robust to $\pm 10\%$ perturbation (Section 4.4), but a different defensible weighting would shift the ranking somewhat. Internal validation (Section 4.3–4.4) shows the method behaves sensibly; it does not field-verify any individual hotspot’s on-the-ground condition — that would need a site visit or independent recent survey.

Engineering. Earth Engine’s ~5-minute interactive timeout is worked around by caching an intermediate asset once, so the first run after any change requires a several-minute export. The dashboard’s map tiles use non-permanent `getMapId()` tokens that need periodic regeneration (a production version should use static tiles or an Earth Engine App). The dashboard was built and tested by the team, not yet reviewed with the agencies who would use it.

Not yet done. A standalone before/after seagrass-change figure, a short demonstration video (optional per the entry guide), and field validation of specific hotspots remain open — tracked openly in the repository.

7. References

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- Landsat 8/9 Collection 2 Level-2 imagery, USGS/NASA, accessed via Google Earth Engine.
- Marine habitat vector data: Environment Agency – Abu Dhabi (EAD).